



Timely ON Switch

Project Report

Course: DLD

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Introduction

A timely ON switch is kind of switch that turns ON for specific amount of time after that specific time its turn off and it stays off till the user input any other time and press the start button or the user press start button with the same previous time. this timely on switch is of importance factories or machine where working of a machine for specific time is very much important. the time switch automatically turn itself off after the inputted time by user.

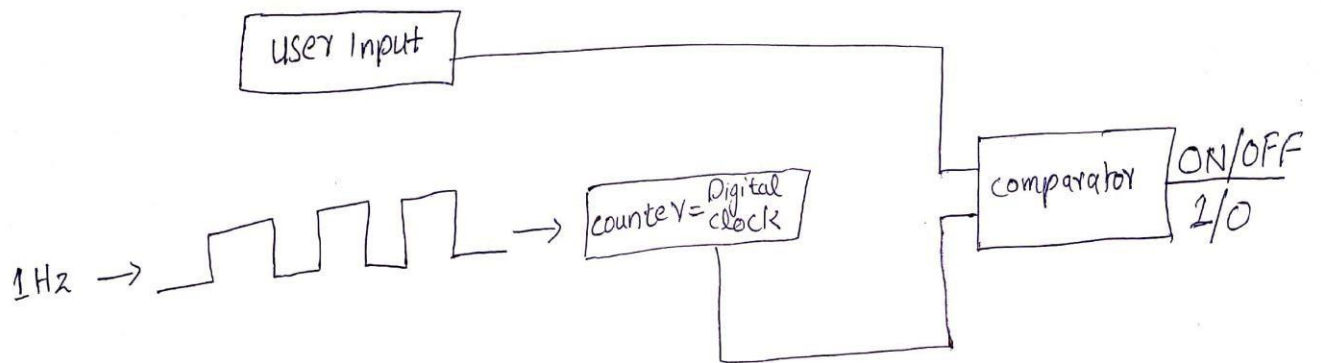
Components

- OR, AND,NOT,Xor gates
- Multiplexer
- Total 19 flip flops
- 12 seven Segment Display
- 1HZ signal clock
- Start and Stop button(logic states)

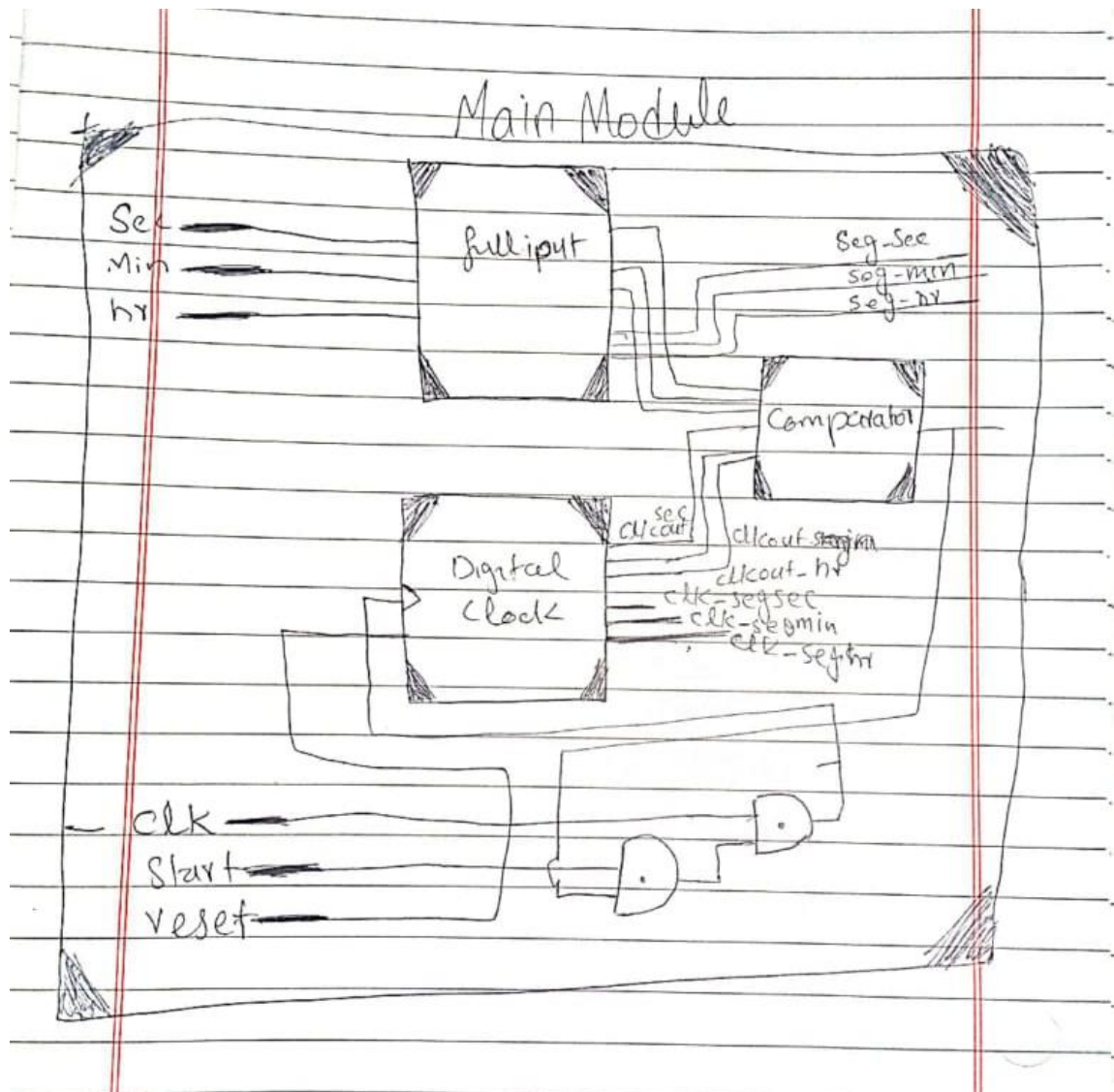
Working of the Switch

The user will first input the time , and then press the start button after pressing start the digital clock made of counters will be started and this clock will be working on 1hz frequency as 1 hz is 1 second, after that the comparator at the end made of XOR gates in this project would give 1 the ON switch but when the time matches with that of user inputted the value at the output of comparator is 0 and the clock stops at that time, the output is 0 hence OFF switch. After that if the user wants to run it again, he will press the reset button and then input another time or same time and then press the start button again. He can input different or same time again. Input will be given in BCD codes. The output is also in bcd codes as well in 7segment codes.

Diagram:



∴ Digital clock will be constructed from counters



Construction:

all of the equation are in folder (Equations);

For inputs:

For 1 to 9 inputs

(3) (2) (1) (0)

A	B	C	D	A _{tri}	B _{tri}	C _{tri}	D _{tri}
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	1
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	1
0	1	1	0	0	1	1	0
0	1	1	1	0	1	1	1
1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	1
1	0	1	0	0	0	0	0
1	0	1	1	0	0	0	0
1	1	0	0	0	0	0	0
1	1	0	1	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0

for 1 to 5 inputs					
(6)	(5)	(4)			
A	B	C	A++	B++	C++
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	0	0	0
1	1	1	0	0	0

Input are handled in a perfect way, when a user input a code for the time , and that code is valid then that code is entered through to project. But if the code is invalid the above equation corrects it by making the invalid code 0. Then when it is corrected it works fine with the project.

Invalid codes are when some one input value greater than 9 at unit places in minute and seconds, or when greater than 5 value is input at the tens place in minute and second. The hours has kind of different structure, when the tens place is zero if person input 0 to 9 its correct and if he inputs greater than 9 the it is corrected by equation, but when tens place has 1 , we cannot input greater than 2 value in unit place, to make it right I used a multiplexer and some combinational code ,when tens place is 1 and someone inputs value greater than 2 than then combinational code outputs 1 and that output goes to multiplexer and mux chooses between given input to unit place and the zero, and the correct is outputted, and then is compared with the digitclock output later.

Digitalclock

For counter from 1 to 10

(3)	(2)	(1)	(0)		A _{tt+1}	B _{tt+1}	C _{tt+1}	D _{tt+1}
A	B	C	D					
0	0	0	0		0	0	0	1
0	0	0	1		0	0	1	0
0	0	1	0		0	0	1	1
0	0	1	1		0	1	0	0
0	1	0	0		0	1	0	1
0	1	0	1		0	1	1	0
0	1	1	0		0	1	1	1
0	1	1	1		1	0	0	0
1	0	0	0		1	0	0	1
1	0	0	1		1	0	1	0
1	0	1	0		x	x	x	x
1	0	1	1		x	x	x	x
1	1	0	0		x	x	x	x
1	1	0	1		x	x	x	x
1	1	1	0		x	x	x	x
1	1	1	1		x	x	x	x

for counter from 1 to 6					
(6)	(5)	(4)			
A	B	C	A _{HH}	B _{HH}	C _{HH}
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	x	x	x
1	1	1	x	x	x

the digital clock is made so that there are 3 blocks, 1 block for seconds, 1 for minutes, and 1 for hours. Now 1 hz clock is provided to it, the each block has 2 digit place unit and ten, lets start with the seconds block, when the unit digit place value reach 9, after another clock cycle it will be 10 according to above code, this 10 occurring cause this unit place digit to reset to zero, and provide a positive edge pulse to the tens digit place making it to change to 1 from 0, after another 9 at unit place, after will occur 10 which will reset the unit place again to zero and providing positive edge pulse to tens place and making it 2 so on it goes. When the tens place have 5 its value and unit place have 9 its value, now when another cycle occurs, the 10 state will arrive which will reset the digits place, and provide positive edge to tens place changing it to 6, and occurring of 6 at tens reset the tens place to 0, and provides a positive edge pulse to the next unit digit place of minute. The working of minutes is same as seconds. The hours, however, has one different thing it made of 5 bits, 1 for tens digit place 4 for unit digit place, now the hours works the same when 10 arrive at unit digit place it reset unit place and changes tens place from 0 to 1, but when the tens place is 1 we cannot have value greater than 2 at the unit place, so when the value at tens is one and at units gets greater than 2 we reset both of the unit and tens digit place to start 00 hours again

it makes possible the time 12:59:59 to 00:00:00 when the clock cycle occurs.

Note: the clock is only for 12 hours , greater than 12 hours cannot be inputted.

Comparator:

Comparator is simple the valid input that comes from fullinput and the output of clock is compared, corresponding bits are compared with each other when they do not match the output is 1(on switch), when they match the output gets zero and also stops the clock there.

Misc.

Start and the output of the project are AND together , and the output of this AND is AND together with the clock make it possible for the clock to stop when start is zero or when the output of the project gets to zero.

Usage :

1) Various Machineries

The above mention switch is essential for machineries that require perfect timming for a task, they can also be combined with other logic and turn ON this switch for specific time inputted from that logic.

2) Alarm Clocks

If we inverse the output it turns into an Alarm clock that can be used for several task in factories or in daily life.

3) With other switch:

Through simple logic can be combined with another normal switches to make it into both normal and timely ON switch.

Conclusion:

This project is time switch that turn off with time inputted by user, digital clock ,comparator and input module are used in this. The user input a time the switch turn off after that time, it can be used in a lot of ways, some are explained above.

I have learned a lot creating this project , I learned how to built a proper project, moreover many of my doubts were cleared of using different logic gates, ics etc,

I started this project with simple input then tried more and more introduced correction to it. I converted the counter and made a full digital clock with it,plus the

working of comparator. The code cleared my mind of how things flow in digital logic. how different modules are interconnected with each other to make a project, and moreover how can I make a better project I understood all. It was an interesting project